Optics – from Fermat's principle to wave optics

Optics is the branch of physics in which the battle over the nature of light was fought: rays or particles, waves or photons. The answer turned out to be "all of them, depending on the question". This page follows a university course in optics: geometrical optics with Fermat's principle and refraction, wave optics with interference and diffraction, Fresnel's equations for reflection and polarisation, and finally resolving power – the limit of what a microscope or telescope can see. The site's interactive lens page and Fourier optics page go further with image formation.

The dispute over the nature of light

Around 1020, in his *Book of Optics*, Ibn al-Haytham (Alhazen) showed that we see because light travels from the object into the eye – not the other way round, as the Greek philosophers had believed. Willebrord Snell found the law of refraction in 1621, and Descartes published it in 1637. In 1690 Christiaan Huygens explained refraction and reflection with waves, while Newton in his *Opticks* (1704) preferred particles – light travels in straight lines, whereas waves bend round corners. Newton's authority settled the matter for a hundred years.

In 1801 Thomas Young demonstrated interference from two slits. In 1818 Augustin Fresnel gave a complete wave theory of diffraction. Siméon Poisson, an opponent, worked out that the theory predicted a bright spot in the middle of the shadow of a round disc – surely absurd. François Arago did the experiment and found the spot. Since 1818 it has been called Arago's (or Poisson's) spot. In 1865 Maxwell showed what kind of waves they are (see Electromagnetism), and in 1905 Einstein brought the particles back as photons (see Quantum mechanics).

Fermat's principle

In 1662 Pierre de Fermat formulated the basic principle of optics: light follows the path between two points for which the optical path length – and therefore the time – is stationary (usually a minimum):

$$\delta \int n(\vec{r})\, ds = 0, \qquad n = \frac{c}{v}$$

Both the law of reflection and Snell's law follow from this. Fermat's principle is the ancestor of the variational principles that later became the foundation of all of mechanics (see Least action) – and Feynman showed that in quantum mechanics it results from all paths contributing, but only

the stationary ones interfering constructively. A mirage above a hot road is Fermat's principle in practice: the light takes a curved path through air with a varying refractive index.

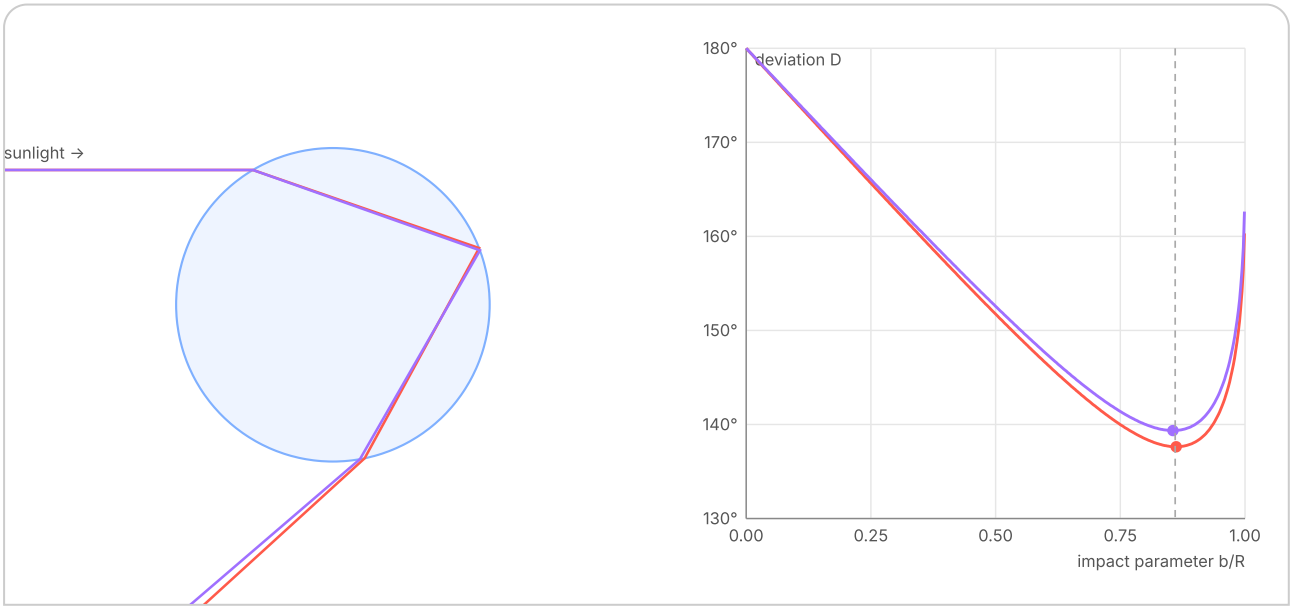
Refraction and total internal reflection

$$n_1 \sin\theta_1 = n_2 \sin\theta_2, \qquad \sin\theta_c = \frac{n_2}{n_1} \quad (n_1 > n_2)$$

When light passes from a denser to a less dense medium and the angle of incidence exceeds the critical angle θ_c, all the light is reflected – total internal reflection. For glass to air $\theta_c \approx 42^\circ$, for diamond only 24°, which gives diamonds their sparkle. In an optical fibre the light is trapped by total internal reflection over thousands of kilometres. In 1966 Charles Kao showed that glass could be made pure enough for this, and received the Nobel Prize in 2009. The refractive index depends on wavelength (dispersion), so a prism spreads white light into colours – Newton's famous experiment of 1666.

The rainbow

In 1637 Descartes explained the rainbow by following rays through a water drop: the light is refracted on entry, reflected once at the back and refracted on exit. The angle of deviation depends on where the ray strikes the drop, but has a minimum of about 138°. At the minimum many rays bunch together – that is the rainbow, which is therefore seen 42° from the point directly opposite the Sun. Because red light is refracted slightly less than violet, red is on the outside. The secondary bow (two reflections) lies at about 51° with the colours reversed, and the sky between the two bows is darker: Alexander's dark band.



Lenses and matrix optics

A thin lens with radii of curvature R_1 and R_2 has a focal length given by the lensmaker's formula, and the image is found with the lens formula:

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right), \qquad \frac{1}{a} + \frac{1}{b} = \frac{1}{f}$$

In paraxial optics (small angles) a ray is described by its height y and angle θ, and each optical element is a 2×2 matrix. A whole system – a pair of binoculars, a microscope or a laser resonator – is the product of the matrices:

$$\begin{pmatrix} y_2 \\ \theta_2 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} y_1 \\ \theta_1 \end{pmatrix}, \qquad \text{free space: } \begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}, \quad \text{thin lens: } \begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix}$$

A resonator is stable when $|A + D| \le 2$ for a complete round trip (see Laser physics). Real lenses have aberrations: spherical aberration (marginal rays are focused too close), chromatic aberration (dispersion), coma and astigmatism. Ludwig Seidel classified them in 1857. In 1990 the Hubble telescope's mirror turned out to have been ground 2.2 µm out and gave spherical aberration – corrected with corrective optics in 1993.

Interference

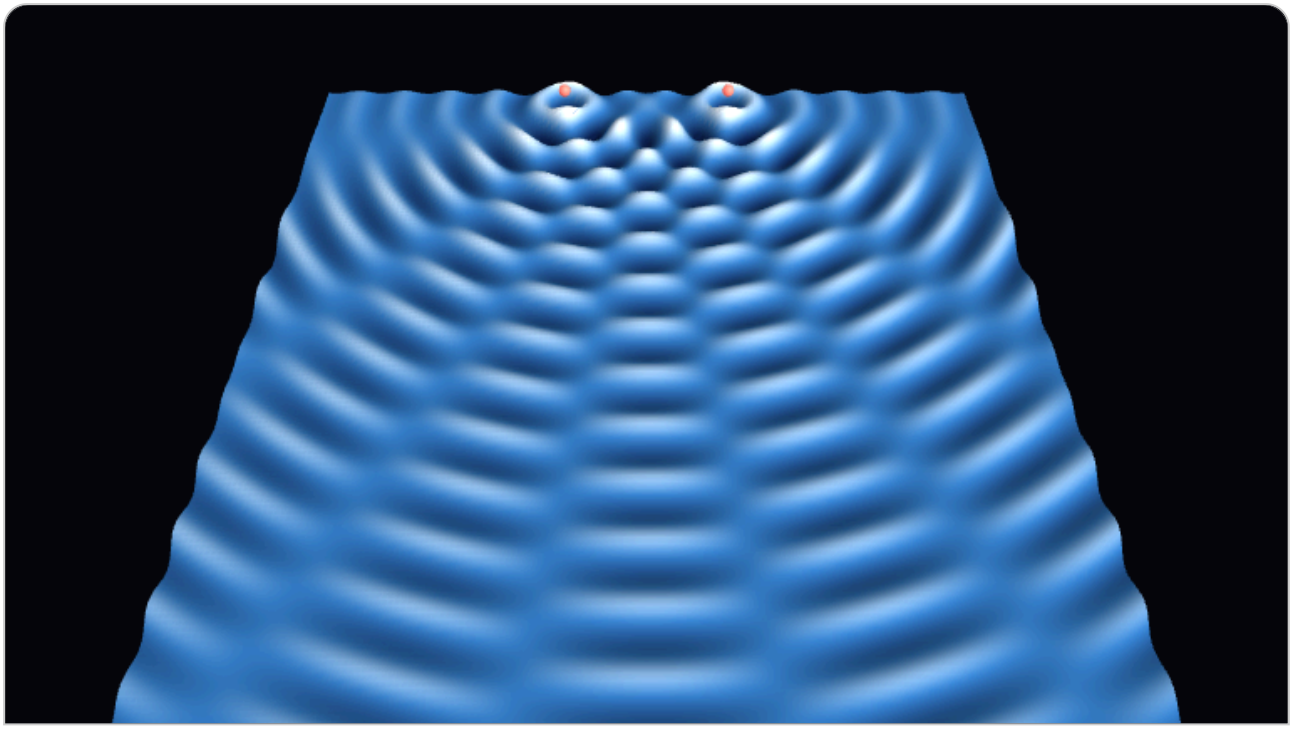
When two coherent waves meet, it is the fields that add – not the intensities. With phase difference δ:

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\delta, \qquad \delta = \frac{2\pi}{\lambda}\,\Delta s$$

In Young's double slit with spacing d the maxima lie at $d\sin\theta = m\lambda$. Thin films – soap bubbles, oil on water, anti-reflection coatings on spectacles – are coloured by interference between light reflected from the top and bottom surfaces. In 1881–87 Albert Michelson's interferometer split light into two arms; with Edward Morley he found no difference in the speed of light in different directions, and no ether (see Relativity). In 2015 the same principle, with arms 4 km long, enabled LIGO to measure gravitational waves: a change in arm length of 10^{-18} m.

Interference requires *coherence*. Sunlight has a coherence length of about 1 µm, a laser can have kilometres. That is why colours are seen only in a thin soap film, not in a thick sheet of glass.

Interference in a ripple tank (3D)

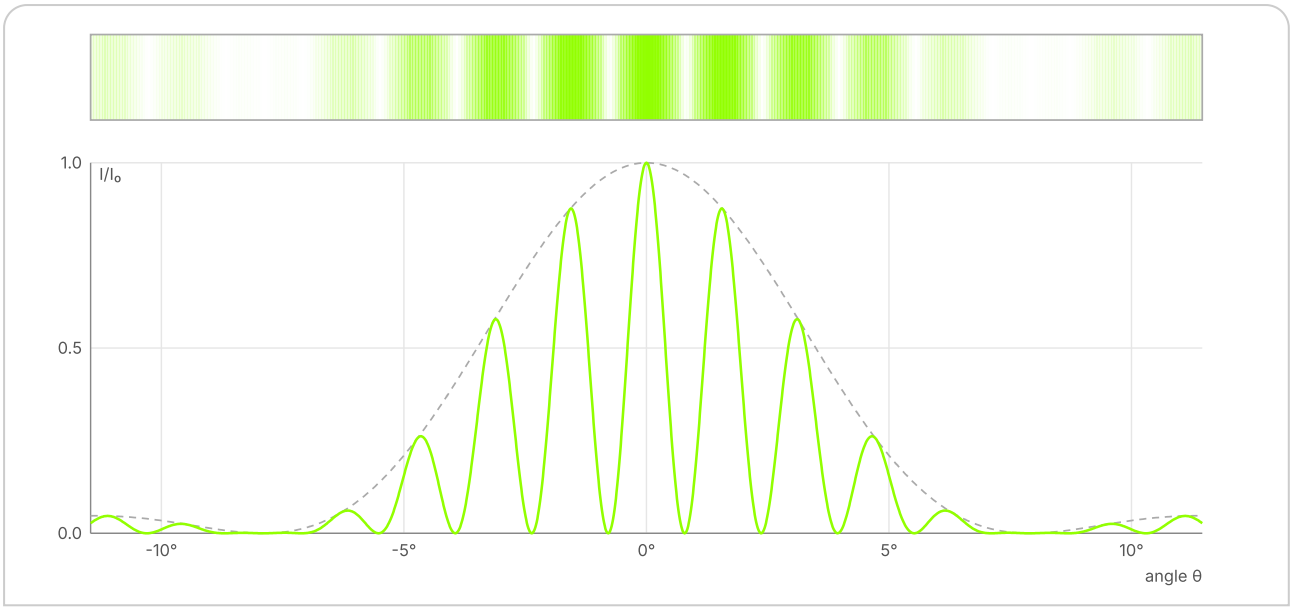


Diffraction

The Huygens–Fresnel principle: every point on a wavefront is a source of new spherical wavelets, and the wave field further on is the sum of them all. Far from an aperture (Fraunhofer diffraction) the pattern is the Fourier transform of the aperture (see The lens as a Fourier transformer and Fourier). For N slits of width a and spacing d:

$$I(\theta) = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \left(\frac{\sin N\alpha}{N\sin\alpha}\right)^2, \qquad \beta = \frac{\pi a \sin\theta}{\lambda}, \quad \alpha = \frac{\pi d \sin\theta}{\lambda}$$

The first factor is the single-slit pattern (the envelope), the second the interference between the slits. With many slits – a diffraction grating – the principal maxima become very sharp, and the grating separates colours far better than a prism. Joseph von Fraunhofer built the first good gratings and in 1814 found the dark lines in the solar spectrum that became the key to determining the composition of the stars (see The spectrum). The resolving power is $\lambda/\Delta\lambda = mN$.

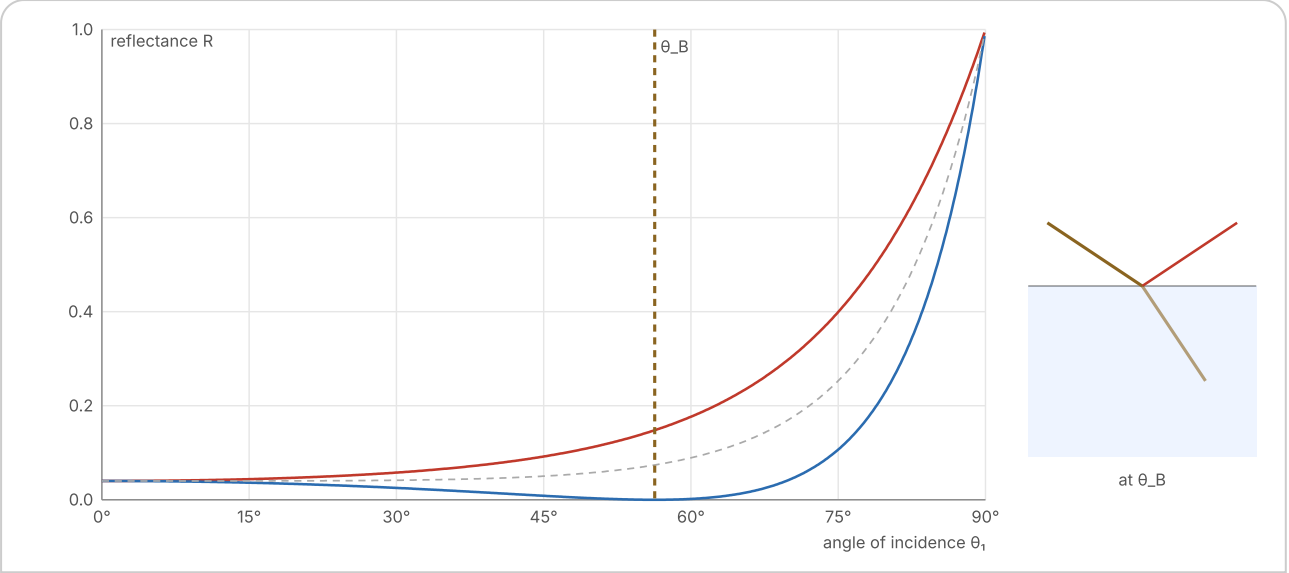


Fresnel’s equations and polarisation

How much light is reflected from a glass surface? Fresnel derived it in 1823 – before Maxwell – from boundary conditions on the waves. The answer depends on the polarisation: s (field perpendicular to the plane of incidence) and p (in the plane of incidence):

$$r_s = \frac{n_1\cos\theta_1 - n_2\cos\theta_2}{n_1\cos\theta_1 + n_2\cos\theta_2}, \qquad r_p = \frac{n_2\cos\theta_1 - n_1\cos\theta_2}{n_2\cos\theta_1 + n_1\cos\theta_2}, \qquad R = |r|^2$$

At normal incidence the fraction $\left(\frac{n_1-n_2}{n_1+n_2}\right)^2$ is reflected – 4 % for glass. At Brewster’s angle $\tan\theta_B = n_2/n_1$ (David Brewster, 1815) no p-polarised light at all is reflected; the reflected light is completely polarised. That is why polarising sunglasses remove glare from water and roads (see Polarisation). Étienne-Louis Malus discovered polarisation by reflection in 1808, when he looked at the sunset reflected in the windows of the Luxembourg Palace through a calcite crystal.



Resolving power

Even a perfect lens cannot image a point as a point: diffraction at the circular aperture gives an Airy disc (George Airy, 1835). In 1879 Lord Rayleigh defined two points as just resolved when the maximum of one lies on the first zero of the other:

$$\theta_{\min} = 1.22\,\frac{\lambda}{D}, \qquad d_{\min} \approx \frac{0.61\,\lambda}{n \sin\alpha}\ \text{(Abbe, microscope)}$$

The pupil of the eye (3 mm) gives about 1 arcminute. A 2.4 m mirror like Hubble's gives 0.05″, but on the ground air turbulence ("seeing") limits it to about 1″ – unless adaptive optics is used. In 1873 Ernst Abbe derived the limit of the microscope at about half a wavelength. It was broken with fluorescence microscopy (STED, PALM), which earned Eric Betzig, Stefan Hell and William Moerner the Nobel Prize in Chemistry in 2014. In 2019 the Event Horizon Telescope achieved the resolution of an Earth-sized antenna and imaged a black hole.

Modern optics

Dennis Gabor invented holography in 1947 – a recording of both amplitude and phase. It only became practical with the laser (see Laser physics). Non-linear optics, in which strong fields double the frequency (Franken, 1961), gives green laser pointers. Optical frequency combs (Hall and Hänsch, Nobel Prize 2005) measure frequencies to 18 digits and are used in atomic clocks. Metamaterials with negative refractive index, photonic crystals and quantum optics with single photons – quantum communication and cryptography – are active research fields today. And glass fibres carry by far the greater part of the world's internet traffic.

Further reading

- Eugene Hecht: *Optics* (5th ed., Pearson 2017)
- Max Born and Emil Wolf: *Principles of Optics* (7th ed., Cambridge 1999)
- Richard Feynman: *QED. The Strange Theory of Light and Matter* (Princeton 1985)
- Richard Feynman: *The Feynman Lectures on Physics*, vol. I, ch. 26–33

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